

SIMATS ENGINEERING

ARTIFICIAL INTELLIGENCE (CSA17) ASSESSMENT TOOL 1 — ANALYTICAL PROBLEM SOLVING

Question 1: Water Jug Problem

Question: You are given two jugs, one with a capacity of 4 gallons and another with a capacity of 3 gallons. Neither jug has measuring marks. Using a water pump, how can you obtain exactly 2 gallons in the 4-gallon jug? Assume that a jug can be filled, emptied onto the ground, or poured into the other jug, and no other measuring devices are available.

Answer: Represent each state as (x, y) , where x is the amount of water in the 4-gallon jug and y is the amount in the 3-gallon jug.

Step	Action	State (4-gallon, 3-gallon)
1	Start with both jugs empty.	(0, 0)
2	Fill the 4-gallon jug.	(4, 0)
3	Pour from the 4-gallon jug into the 3-gallon jug until the 3-gallon jug is full.	(1, 3)
4	Empty the 3-gallon jug.	(1, 0)
5	Pour the remaining 1 gallon into the 3-gallon jug.	(0, 1)
6	Fill the 4-gallon jug again.	(4, 1)
7	Pour into the 3-gallon jug until it is full. It needs 2 more gallons.	(2, 3)

Concept: This is a state-space search problem. The state records the amount of water in each jug, actions change the state, and the goal test checks whether the 4-gallon jug contains exactly 2 gallons.

Conclusion: The goal state is $(2, 3)$, so the 4-gallon jug contains exactly 2 gallons.

Question 2: Mars Rover as an Intelligent Agent

Question: Explain how a Mars Rover analyzes and explores samples on Mars by answering: (i) What are the percepts for this agent? (ii) Characterize the operating environment. (iii) What actions can the agent take? (iv) How can its performance be evaluated? (v) What agent architecture is most suitable, and why?

Answer:

(i) Percepts: Images from cameras; temperature and environmental readings; proximity and navigation sensor readings; and scientific measurements from instruments such as spectrometers used to analyze soil and rock samples.

(ii) Operating environment: The rover operates on Mars. It is partially observable because sensors provide incomplete information; stochastic because terrain and equipment conditions are uncertain; sequential because current actions affect later decisions; dynamic because conditions can change; continuous because movement and measurements can take continuous values; and single-agent when the rover alone is considered the decision-maker.

(iii) Actions: Move and turn; capture images; avoid obstacles; select routes; collect soil and rock samples; drill into rocks; analyze samples; and transmit data to Earth.

(iv) Performance evaluation: Scientific accuracy, number and quality of samples collected, useful data transmitted, navigation success, obstacle avoidance, energy efficiency, mission completion, and protection of the rover and instruments.

(v) Suitable architecture: A model-based, goal-based agent with utility-based decision-making is suitable. The model-based component maintains information about terrain and obstacles; the goal-based component plans how to reach locations and collect samples; and the utility-based component compares choices by safety, scientific value, time, and energy.

Concept (PEAS): Performance measure—scientific discoveries, safety, and energy efficiency; Environment—Martian terrain and conditions; Actuators—wheels, robotic arm, and drill; Sensors—cameras, spectrometers, and temperature sensors.

Question 3: 8-Queens Problem

Question: Formulate the problem components for placing eight queens on a chessboard so that no queen attacks any other queen. Explain how search strategies can explore the state space.

Answer:

- 1. Initial state:** An empty 8×8 chessboard.
- 2. State space:** All possible partial and complete arrangements of eight queens on the board.
- 3. Actions:** Place a queen in a safe row of the next column.
- 4. Transition model:** The state changes by placing one queen in the next column without attacking previously placed queens.
- 5. Goal test:** Eight queens are placed and no two share the same row, column, or diagonal.
- 6. Path cost:** If each placement has cost 1, a complete solution requires eight placements.
- 7. Constraints:** No two queens may share a row, column, or diagonal. For queens at (r_1, c_1) and (r_2, c_2) , a diagonal conflict occurs when $|r_1 - r_2| = |c_1 - c_2|$.

Search strategy: Depth-first search with backtracking can be used. Place a queen in a safe position in each column. If no safe position exists in the next column, remove the previous queen and try another row. Stop when all eight queens are placed safely.

Concept: Backtracking avoids exploring complete arrangements that already violate a constraint, reducing unnecessary search.

Question 4: OLA Cab Booking — Problem-Solving Agent

Question: A customer uses the OLA application to travel from one location to another and can choose a cab type such as Mini, Micro, Sedan, Shared, or Prime. Identify the type of problem-solving agent and write pseudocode.

Answer: A utility-based problem-solving agent is suitable because it can compare feasible cab and route options using the customer's preferences for cost, travel time, comfort, and availability.

```
BEGIN
  INPUT source_location, destination_location
  INPUT customer_preferences

  GET available_cab_types
  GET feasible_routes, estimated_fares, travel_times

  best_option = NULL
  best_utility = -INFINITY

  FOR each cab in available_cab_types:
    FOR each route in feasible_routes:
```

```

    IF cab is available AND route is feasible:
        CALCULATE fare, travel_time, comfort_score

        IF option satisfies customer_preferences:
            CALCULATE utility from cost, time, comfort

            IF utility > best_utility:
                best_utility = utility
                best_option = (cab, route)
            END IF
        END IF
    END IF
END FOR

IF best_option is not NULL:
    DISPLAY best_option
    CONFIRM booking with customer
    EXECUTE booking
    FOLLOW selected route until destination
ELSE:
    DISPLAY "No suitable cab available"
END IF
END

```

Concept: A goal-based agent focuses on reaching the destination. A utility-based agent compares different ways of reaching it and chooses the option that best matches the customer's preferences.

Question 5: Uniform Cost Search (UCS)

Question: A logistics company wants to find the least-cost path from starting warehouse S to destination G using the weighted graph provided in the assessment. Apply Uniform Cost Search.

Answer: UCS expands the node with the smallest cumulative path cost first. The given edge costs are $S \rightarrow A = 1$, $S \rightarrow G = 12$, $A \rightarrow B = 3$, $A \rightarrow C = 1$, $B \rightarrow D = 3$, $C \rightarrow D = 1$, $C \rightarrow G = 2$, and $D \rightarrow G = 3$.

Step	Expanded node	Cost calculations / result
1	S (0)	$A = 0 + 1 = 1$; $G = 0 + 12 = 12$
2	A (1)	$B = 1 + 3 = 4$; $C = 1 + 1 = 2$
3	C (2)	$D = 2 + 1 = 3$; $G = 2 + 2 = 4$
4	D (3)	G through D = $3 + 3 = 6$; discard because cost 4 is better
5	B (4)	D through B = $4 + 3 = 7$; discard because D already has cost 3
6	G (4)	Goal reached at the lowest cumulative cost

Optimal path: $S \rightarrow A \rightarrow C \rightarrow G$

Total cost: $1 + 1 + 2 = 4$ units.

Conclusion: The least-cost path from S to G is $S \rightarrow A \rightarrow C \rightarrow G$, with total cost 4. UCS finds the optimal path for this graph because the edge costs are non-negative and it selects the lowest cumulative-cost path first.